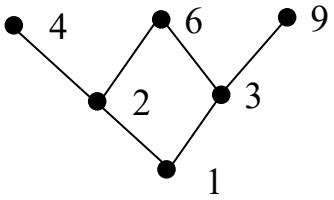
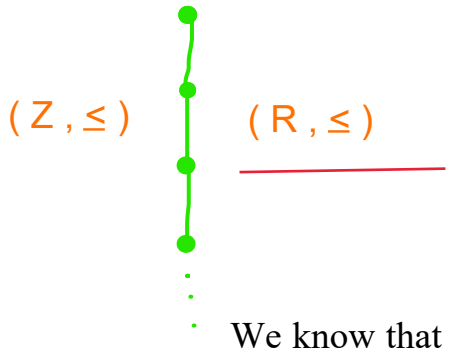
Chapter 1





Partially Ordered Sets

We know that the sets of integers Z and the real numbers R with the natural order ≤ are totally ordered sets, by means that the set of all numbers of **Z** (or of R) can be totally ordered as a set of points on a line or in a chain. So we will say that (**Z**, ≤) and (R, ≤) are totally ordered sets or a chain. Indeed, (**Z**, ≤) and (R, ≤) are special type of partially ordered set.

In fact, we have to begin to define what is a partial order on a set A. A binary relation ≤ on a nonempty set A is said to be a ***partial order*** on A, if ≤ satisfies, for all $x, y, z \in A$,

1. $x \leq x$, (reflexivity)
2. if $x \leq y$ and $y \leq x$, then $x = y$, (antisymmetry)
3. if $x \leq y$ and $y \leq z$, then $x \leq z$. (transitivity)

The pair $(A, \leq)$ is called a ***partially ordered set*** (briefly called a ***poset***).

If the partial order ≤ satisfies in addition the property,

(T). Either $x \leq y$ or $y \leq x$, for all $x, y \in A$,

then the partially ordered set $(A, \leq)$ is called a ***totally ordered set*** (also called a ***linearly ordered set***) or a ***chain.*** I.e. in totally ordered sets each two elements are comparable.

Example 1. Consider the set $A = \{a, b, c\}$, and the relations $\leq_1$ and $\leq_2$ on A as follows: **(i)** $\leq_1 = \{(a, a), (b, b), (c, c), (a, b), (a, c)\}$,

(ii) $\leq_2 = \{(a, a), (b, b), (c, c), (a, b), (b, c), (a, c)\}$.

(i) The system $(A, \leq_1)$ is a poset, because the relations $\leq_1$ is reflexive, Antisymmetric and transitive, but not totally ordered set.

(ii) The system $(A, \leq_2)$ is a totally ordered set, because $(A, \leq_2)$ is a poset and satisfies the property (T).

Example 2. The most natural example of a poset is the power set **P(X),** the collection of all subsets of a set **X**, ordered by the set inclusion $\subseteq$. One can see directly that the system $(\mathbf{P(X)}, \subseteq)$ forms a partially ordered set **(a poset) but** not totally (if **X** has more than one element)

Example 3. Let **N** be the set of all natural numbers $\{0, 1, 2, \ldots\}$ and let $m \leq n$ means m divides n for $m, n \in \mathbf{N}$. Then $(\mathbf{N}, \leq)$ is a poset, but not totally ordered set.

Note that a poset is not totally ordered set if it has two not comparable elements. In the last example we have : $2, 3 \in \mathbf{N}$ and $2 \nleq 3$ and $3 \nleq 2$; i.e. 2 , 3 neither divides the other.

We say that x is covered by y in the poset $(A, \leq)$, written by $x \prec y$

$\nexists$

If $x < y$ and there is no $z \in A$ with $x < z < y$.

In the mathematical discipline known as order theory, a **Hasse diagram**, named after Helmut Hasse (1898–1979)) is a simple picture of a finite partially ordered set, forming a drawing of the transitive reduction of the partial order. Concretely, to define a Hasse diagram for a finite poset $(A, \leq)$, the elements of A are represented by points in the plane and a line is drawn from x up to y precisely when $x \prec y$. Now we can generate lots of examples of posets using Hasse diagram as we will see.

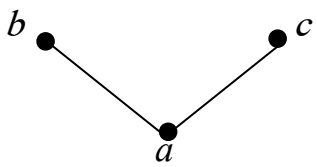
Example 4. The Hasse diagram of the posets $(A, \leq_1)$ and $(A, \leq_2)$ given in example 1 can be described as follows:

N_5

A = { a , b , c }

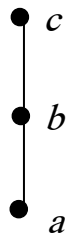
(i) The Hasse diagram of the poset $(A, \leq_1)$ is

(a , a) , (b , b) , (c , c) , (a , b) , (a , c)

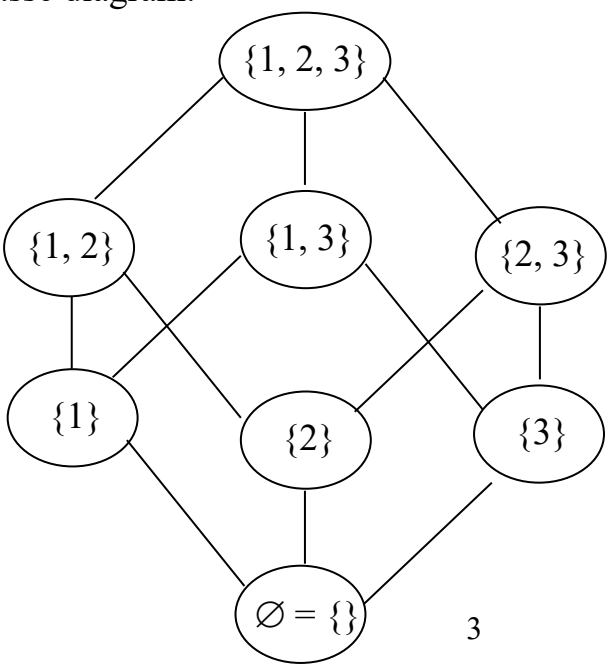


(ii) The Hasse diagram of the totally ordered $(A, \leq_2)$ is

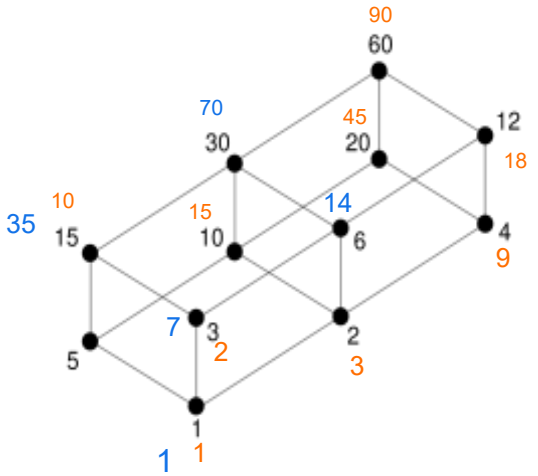
(a , a) , (b , b) , (c , c) , (a , b) , (b, c) , (a , c)



Example 5. The power set of {1, 2, 3} partially ordered by inclusion, has the Hasse diagram:



Example 6. The set A = { 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60 } of all divisors of 60, partially ordered by divisibility, has the Hasse diagram:

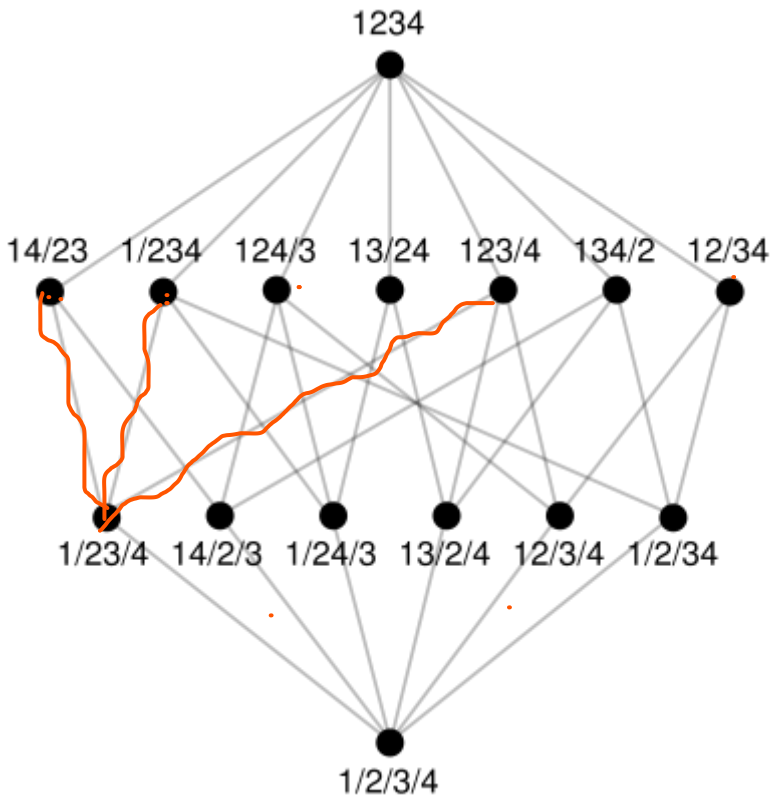


60	1
30	2
20	3
15	4
12	5
10	6

90	1
45	2
30	3
18	5
15	6
10	9

70	1
35	2
14	5
10	7

Example 7. The set of all 15 partitions of the set { 1, 2, 3, 4 }, partially ordered by "refinement", i.e. a finer partition is "less than" a coarser partition, has the Hasse diagram:



{1234}

{12}{34}
{13}{24}
{14}{23}
{1}{234}
{2}{341}
{3}{241}
{4}{123}

{12}{3}{4}
{13}{2}{4}
{14}{2}{3}
{23}{1}{4}
{24}{1}{3}
{34}{1}{2}

{1}{2}{3}{4}

Relations . A relation on a set X is a nonempty subset of X^2. If ρ is a relation on X, then the relation ρ^{-1} is the ***dual relation*** of ρ; i.e.,

$(x_2, x_1) \in \rho^{-1} \Leftrightarrow (x_1, x_2) \in \rho$. This means that $(\rho^{-1})^{-1} = \rho$.

The relation $\Delta = \{(x, x): x \in X\}$ is called the ***Diagonal Relation*** (or ***equality relation***) on X. We have the following facts on the relations:

- A relation ρ is reflexive on $X \Leftrightarrow \Delta \subseteq \rho$.
- A relation ρ is symmetric on $X \Leftrightarrow \rho = \rho^{-1}$.
- A relation ρ is antisymmetric on $X \Leftrightarrow \rho \cap \rho^{-1} \subseteq \Delta$.
- A relation ρ on a set X is called total: $\Leftrightarrow \rho \cup \rho^{-1} = X^2$.
- ρ is equivalent (reflexive, symmetric and transitive) and total on X $\Leftrightarrow \rho = X^2$. (the relation X^2 is denoted usually by ∇).

The **composition** $\rho \circ \sigma$ (or the ***product***) of two relations ρ and σ on a set X is defined by:

$$\rho \circ \sigma := \{ (x, y) \in X^2 : \exists\, z \in X \text{ such that } (x, z) \in \sigma \text{ and } (z, y) \in \rho \}.$$

This means that if ρ and σ are functions on X, then $\rho \circ \sigma$ is the usually composition function; i.e. $\rho \circ \sigma (x) = \rho (\sigma (x))$. We may easily deduce that:

1- • $\rho \circ \Delta = \rho$ and $\Delta \circ \rho = \rho$. (1)

$(x, y) \in \rho \circ \Delta \Leftrightarrow (x, x) \in \Delta \;\&\; (x, y) \in \rho \Leftrightarrow (x, y) \in \rho$. So $\rho \circ \Delta = \rho$.

2- • $\rho \circ \sigma \supseteq \rho$, if $\sigma \supseteq \Delta$, and $\rho \circ \sigma \supseteq \sigma$, if $\rho \supseteq \Delta$. (2)

$(x, y) \in \rho \Rightarrow (x, x) \in \sigma \;\&\; (x, y) \in \rho \Rightarrow (x, y) \in \rho \circ \sigma \Rightarrow \rho \subseteq \rho \circ \sigma$.

3- • A relation ρ is transitive on $X \Leftrightarrow \rho \circ \rho \subseteq \rho$. (3)

$(x, y) \in \rho \circ \rho \Rightarrow \exists\, z : (x, z) \in \rho \;\&\; (z, y) \in \rho \Rightarrow (x, y) \in \rho$ if ρ is transitive. $\Rightarrow \rho \circ \rho \subseteq \rho$.

$(x, z) \in \rho \;\&\; (z, y) \in \rho \Rightarrow (x, y) \in \rho \circ \rho \Rightarrow (x, y) \in \rho \Rightarrow \rho$ is transitive.

4- • A relation ρ is transitive and reflexive on $X \Leftrightarrow \rho \circ \rho = \rho$. (4)

ρ is transitive on $X \Leftrightarrow \rho \circ \rho \subseteq \rho$. And $\rho \circ \sigma \supseteq \rho$, if $\sigma \supseteq \Delta$. But ρ is reflexive $\Rightarrow \rho \circ \rho \supseteq \rho$, because ρ is reflexive. $\Rightarrow \rho \circ \rho = \rho$.

Example 8. Consider the set $X = \{1, 2, 3, 4\}$, and the relations

$\rho = \{(1, 2), (2, 4), (3, 1)\}$, $\sigma = \{(2, 1), (1, 4), (2, 4)\}$, and

$\delta = \{(1, 1), (2, 2), (3, 3), (4, 4), (2, 3)\}$. We have:

1. $\rho \circ \sigma = \{(2, 2)\}$, 2. $\sigma \circ \rho = \{(1, 1), (1, 4), (3, 4)\}$,

3. $\rho \circ \delta = \{(1, 2), (2, 4), (3, 1), (2, 1)\}$,

4. $\rho \circ \rho = \{(1, 4), (3, 2)\}$,

5. $\sigma \circ \sigma = \{(2, 4)\}$,

6. $\delta \circ \sigma = \{(2, 1), (1, 4), (2, 4)\}$,

7. $\delta \circ \delta = \{(1, 1), (2, 2), (3, 3), (4, 4), (2, 3)\}$,

8. $\rho \circ \Delta = \rho$ and $\rho \circ \delta \supseteq \rho$, if $\delta \supseteq \Delta$.

Lcture

Definitions 1. Let $(A, \leq)$ be a poset. An element $m \in A$ is said to be:

(i) a ***maximal*** element in A : $\Leftrightarrow$ " Any element $a \in A$ with $a \geq$ m $\Rightarrow$ $a = m$ ", this means that $m \in A$ is maximal iff there is no element $y \in A$ with $y \geq m$ and $y \neq m$.

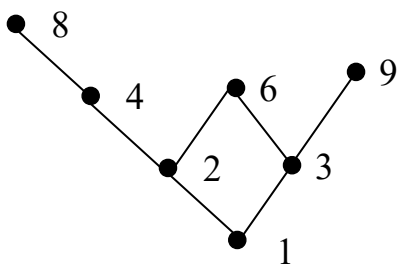
(ii) a ***minimal*** element in A : $\Leftrightarrow$ " Any element $a \in A$ with $a \leq m \Rightarrow$ $a = m$ ", this means that $m \in A$ is minimal iff there is no element $y \in A$ with $y \leq m$ and $y \neq m$.

(iii) a ***largest*** element in A : $\Leftrightarrow$ " $m \geq a$ for all $a \in A$ ", this means that $m \in A$ is a largest in A iff m is greater than any other element in A

(iv) a ***smallest*** element in A : $\Leftrightarrow$ " $m \leq a$ for all $a \in A$ ", this means that $m \in A$ is a smallest in A iff m is leas than any other element in A.

Example 9. Let $S = \{1, 2, 3, 4, 6, 8, 9\}$ and let $m \leq n$ mean that m divides n. It is easy to see that $(S, \leq)$ forms a poset. The Hasse diagram of $(S, \leq)$ is:

(S , ≤)



(i) 6, 8, and 9 are maximal elements in $(S, \leq)$.

(ii) 1 is the only minimal element in $(S, \leq)$.

(iii) There is no largest element in $(S, \leq)$.

(iv) 1 is the smallest element in $(S, \leq)$.

Example 10. The power set $\mathrm{P}(A)$ of the set $A = \{x, y, z\}$ is partially ordered set by inclusion $\subseteq$.

(i) A is the largest element and the unique maximal element in $(\mathrm{P}(A), \subseteq)$.

(ii) $\varnothing$ is the smallest element and the unique minimal element in $(\mathrm{P}(A\ , \subseteq)$.

Example 11. Let $\mathbf{Eq}(X)$ be the set of all equivalent relations on X. $\mathbf{Eq}(X)$ with the inclusion $\subseteq$ forms a poset.

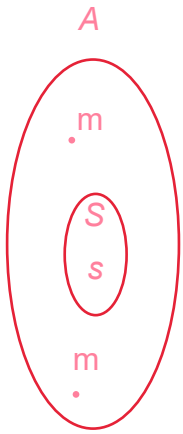
- The Diagonal Δ on X is the smallest equivalent relation on X (Δ is the unique minimal relation in $\mathbf{Eq}(X)$).
- $\nabla := X^2$ is the largest equivalent relation on X (∇ is the unique maximal element in $(\mathbf{Eq}(X), \subseteq)$). Note that $\nabla := X^2$ is not poset.
- The intersection of two equivalent relations is an equivalent relation.

(29/10/2021)

Theorem 1. *Let $(A, \leq)$ be a poset. There is at most one largest element in A. If it exists, it is the unique maximal element in A. Similarly, there is at most one smallest element in A. If it exists, it is the unique minimal element in A.*

Proof. Let m and $m`$ be two largest elements in $A$. Then we have $m \leq m`$ and $m` \leq m$, which implies that $m = m`$. Hence the largest element is unique. Similarly, the smallest element is unique.

Let $(A, \leq)$ have a largest element m. Assume that the poset $(A, \leq)$ has a the maximal element m^*; i.e., if $a \in A$ with $a \geq m^*$, then $a = m^*$. Since m is m is largest element in A, it follows that $m \geq m^*$. Hence, $m = m^*$, therefore, the unique maximal element in A. Similarly, the smallest element is the unique minimal element in A. □ *a*



.**Definitions 2.** Let $(A, \leq)$ be a poset and S be a subset of A. Let $m \in A$

(i) m is a ***lower bound*** of S : $\Leftrightarrow$ $m \leq s$; for all $s \in S$.

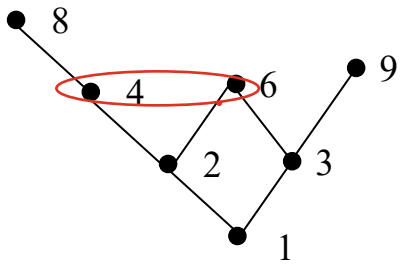
(ii) m is a ***upper bound*** of S : $\Leftrightarrow$ $s \leq m$; for all $s \in S$.

(iii) If m is a lower bound of S and greater than or equal to every lower bound of S, then m is called the ***greatest lower bound*** of S (or briefly **GLB**(S)). Also, m is called the ***infimum*** of S (or briefly **Inf** (S)) if m is the greatest lower bound of S.

(iv) If m is an ***upper bound*** of S and less than or equal to every upper bound of S, then m is called the ***least upper bound*** of S (or briefly **LUB**(S)). Also, m is called the ***supremum*** of S (or briefly **Sup** (S)) if m is the least upper bound of S.

Example 12. In Example 8**,** we have the set $A = \{1, 2, 3, 4, 6, 8, 9\}$ and $m \leq n$ means that m divides n. The Hasse diagram of the poset $(A\ ,\ \leq)$ is:

S = { 2, 4, 6 }



(i) Let $S_1 = \{4, 6\}$, the numbers 2 and 1 are lower bounds of S_1;

i.e., the set of all lower bounds of S_1 is $\{1, 2\}$ or **LB**$(S_1) = \{1, 2\}$.

The set of lower bounds to the subset $S_2 = \{4, 6, 9\}$ is the subset

LB$(S_2) = \{1\}$.

(ii) Also, we may say that 2 is the greatest lower bound of S_1 or **GLB**$(S_1) = 2$, and 1 is the greatest lower bound of S_2 or

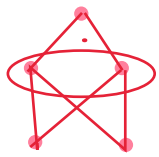
GLB$(S_2) = 1$,

(iii) There is no upper bounds neither of the subset $S_1 = \{4, 6\}$ nor the subset $S_2 = \{4, 6, 9\}$. Let $S_3 = \{1, 4\}$, the numbers 4 and 8 are upper bounds of S_3; i.e., the set of all upper bounds of S_3 is $\{4, 8\}$ or **UB**$(S_3) = \{4, 8\}$.

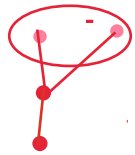
(iv) The **LUB**(S_1) and **LUB**(S_2) are not exist, but the **LUB**(S_3) exists and equal to 4.

In general we note that:

- If $a \leq b$ in a poset $(A, \leq)$, then $b =$ **LUB**$(\{a, b\})$ and $a =$ **GLB**$(\{a, b\})$.



- For a subset S of a poset $(A, \leq)$, the **LB**(S) and the **UB**(S) may be any subset of A, and the **GLB**(S) and the **LUB**(S) may be or may be not exist.



- For a subset S of a poset $(A, \leq)$, if the **GLB**(S) or the **LUB**(S) exists, then **GLB**(S) or **LUB**(S) may be an element in S or may be not.

P(A)

Example 13. In example **5,** we have the power set $\mathcal{P}(A)$ of the set $A = \{x, y, z\}$ partially ordered by inclusion $\subseteq$**.** If we consider the subset $S = \{\{x\}, \{y\}\}$ of the poset $(\mathbf{P}(A), \subseteq)$, then **LB**$(S) = \{\varnothing\}$ and the **UB**$(S) = \{\{x, y\}, \{x, y, z\}\}$, and so also **GLB**$(S) = \varnothing =$ **Inf**(S) and the **LUB**$(S) = \{x, y\} =$ **Sup**(S). In

this example, note that the **GLB**(S) and the **LUB**(S) for each 2-element subset S of the poset ($\mathcal{P}(A)$, $\subseteq$) exist.

Corollary 2. *Let* $(A, \leq)$ *be a poset and* S *be a subset of* A. *Then* S *has at most one supremum* (*one* **LUB**(S)).

Proof. We consider the set **UB**(S) of all upper bounds of S. We have that (**UB**(S), $\leq$) is a poset, then from Theorem 1, (**UB**(S), $\leq$) has at most one smallest element and if it exists, it is unique. Therefore, then S has at most one supremum. □ (P.7)

It is easily see that a similar result with respect to the infimum of S.

Theorem 3. *Let* A *be a nonempty finite set and* $(A, \leq)$ *be a poset. Then we have:*

1. *A has at least one maximal element.*
2. *For each* $x \in A$, *there is a maximal element* m *in* $(A, \leq)$ *with* $m \geq x$.
3. *If* A *has a unique maximal element, then it is also the largest element in* A.

XXX **Proof.**

1. Let F be any subset of A, then $(F, \leq)$ is a poset. We will prove that $(F, \leq)$ has a maximal element.

By Math. Induction on $|F|$.

If $|F| = 1$, then $F = \{a\}$, hence a itself is a maximal element of F.

Let $|F| = n$. By induction hypothesis, let F- $\{a\}$ for $a \in F$ have a maximal element m. If m is a maximal in F, then F has already a maximal element. Let m be not maximal in $F \Rightarrow$ there is an element $b \in F$ with $b \geq m$ and $b \neq m$. To prove that $b = a$, assume that $b \neq a \Rightarrow b \in F$- $\{a\}$

and $b \geq m$ & $b \neq m$ which contradicts that m is a maximal element in F-$\{a\}$. Hence $b = a$. This means that m is a maximal element in F-$\{a\}$ and a

$\geq m$ & $a \neq m$. This implies directly that a is a maximal element of F. Therefore, we may say that any subset F of A has a maximal element. Specially, A has at least one maximal element.

2. Let $x \in A$ and G_x ' $\{ y \in A : y \geq x \}$. According to 1, the poset $(G_x, \leq)$ has a maximal element m. We note that m is also a maximal element in $(A, \leq)$, because if there is an element $m` \in A$ with $m` \geq m \Rightarrow m` \in G_x$. Since $m$ is a maximal element in $G_x \Rightarrow m \geq m`$. This means that $m` = m$. Hence there is no $m` \in A$ with $m` \geq m$. This means that m is a maximal element in $(A, \leq)$. Therefore, there is always a maximal element m in A with $m \geq x$ for each $x \in A$.

3. Let m be the unique maximal element in A. If m is not the largest element of A, then there is an element $a \in A$ satisfying that a — m. Consider the subset G_a ' $\{ y \in A : y \geq a\}$. From **1.** G_a has a maximal element $m`$ and in the same time $m`$ is also a maximal element in A. We note that $m \notin G_a$, because a — m. But $m` \geq a$. This means that $m$ and $m`$ are two distinct maximal elements in A which contradicts the assumption that A has a unique maximal element. Therefore, m is the largest element of A. □

According to Theorem 1 and 3, we may say that:

m is the largest element of the poset $(A, \leq) \Leftrightarrow m$ is the unique maximal element in $(A, \leq)$.

The same results can be obtained with respect to the minimal element.

Theorem 3`. *If $(A, \leq)$ is a finite poset, then we have:*

1. *A has at least one minimal element.*
2. *For each $x \in A$, there is a minimal element m in $(A, \leq)$ with $m \leq x$.*
3. *If A has a unique minimal element, then it is also the smallest element in A.*

xxx ==========

Lemma 4. *Let* $(A, \leq)$ *be a poset. The following are equivalent:*

1. *Every nonempty subset* S *contains an element minimal in* S.
2. $(A, \leq)$ *contains no infinite descending chain* $a_0 > a_1 > a_2 > \dots$.
3. *If* $a_0 \geq a_1 \geq a_2 \geq \dots$ *is a chain in* $(A, \leq)$, *then there exists* k *such that* $a_n = a_k$ *for all* $n \geq k$.

Proof. The equivalence of **2** and **3** is clear.

1 ⇒ 2 : If $(A, \leq)$ *contains an infinite descending chain* $a_0 > a_1 > a_2 > \dots$, then the subset $S = \{a_0, a_1, a_2, \dots\}$ has no minimal element.

2 ⇒ 1: Suppose $(A, \leq)$ fails **1**, and that $S \subseteq A$ has no minimal element. Let f be a choice function on the subsets of S; i.e., for each subset $T \subseteq S$, $f(T) \in T$. Let $a_0 = f(S)$ and $a_1 = f(\{s \in S : s < a_0\})$, … , $a_{i+1} = f(\{s \in S : s < a_i\})$. The subset $\{s \in S : s < a_i\}$ is nonempty for each integer i because S has no minimal element. Then we have $a_0 > a_1 > a_2 > \dots$ an infinite descending chain ⇒ hence $(A, \leq)$ fails 2. □

The conditions described by the preceding lemma are called the descending chain condition (**DCC**). So we have the fact that every finite poset satisfies the **DCC**.

If we replace maximal instead of minimal, ascending chain $a_0 < a_1 < a_2 < \dots$instead of the descending chain $a_0 > a_1 > a_2 > \dots$, and the chain $a_0 \leq a_1 \leq a_2 \leq \dots$ instead of $a_0 \geq a_1 \geq a_2 \geq \dots$, we get a similar equivalence for the property which is called ascending chain condition (**ACC**).

Example 14. Let A = the real interval $[0: 1]$, we have that $([0: 1], \leq)$ is a poset with the ordinary order $\leq$. The interval $(0: 1)$ is a subset of $[0: 1]$ and $(0: 1)$ has neither minimal nor maximal element. Moreover, the poset $[0: 1]$ has an infinite descending chain $1 > 1/2 > 1/3 > \dots$. Then we can say that the poset $([0: 1], \leq)$ has no the property **DCC**.

On other side, we may say that every finite subset of [0: 1] has a minimal and a maximal element.

xxxxxxx **Definitions 3.** Let $(A, \leq)$ be a poset.

1. $(A, \leq)$ is called ***directed*** $:\Leftrightarrow$ each 2-elment subset of A has an upper bound.
2. $(A, \leq)$ is called ***inductively ordered*** $:\Leftrightarrow$ every nonempty totally ordered subset of A has an upper bound.
3. $(A, \leq)$ is called ***strictly inductively ordered*** $:\Leftrightarrow$ every nonempty totally ordered subset of A has a least upper bound (**LUB**).

Example 15. In examples 5 and 11, we have:

1. The power set $\mathcal{P}(A)$ for any nonempty set A forms a poset with the inclusion $\subseteq$. The **GLB**(S) and the **LUB**(S) for each 2-element subsets $S = \{S_1, S_2\}$ of the poset $(\mathcal{P}(A), \subseteq)$ exist**.** Then $(\mathcal{P}(A), \subseteq)$ is directed poset. "**GLB**$(S) = S_1 \cap S_2$ and **LUB**$(S) = S_1 \cap S_2$

   While the poset given in example 9 is inductively ordered , but not directed.

   In fact, the power set $(\mathcal{P}(A), \subseteq)$ is also strictly inductively ordered.

2. Indeed, every finite poset is inductively ordered. Moreover, every finite poset is strictly inductively ordered. For instance, the example given in example 9 is a strictly inductively ordered set.

3. The real interval (0: 1) is a poset with. the usually order $\leq$, but not inductively ordered.

Lemma 5. (Zern's Lemma).

Let $(A, \leq)$ *be a nonempty inductively poset.Then there exists a maximal element in* A.

Exercise 1

1. Find a relation on the set $X = \{x, y, z\}$ which is neither symmetric nor antisymmetric!
2. Find: (a) all equivalent relations on the set $X = \{x, y, z\}$ and (b) all partial order relations on a 3-element set!
3. Show that if ρ is a poset on a set X, then ρ^{-1} is also a poset.
4. Let ρ, σ and δ be any three relations on the set A.

   Prove that $(\rho \cap \sigma) \circ \delta = \rho \circ \delta \cap \sigma \circ \delta$.
5. Prove that the intersection of two posets on a set A is a poset and the intersection of two equivalent relations on a set A is an equivalent relation!
6. Let ρ and σ be reflexive relations on the set A. Show that $\rho \circ \sigma$ is reflexive containing $\rho \cup \sigma$.
7. Consider the set $X = \{a, b, c\}$, find the following relations on X: (1) Totally ordered set. (2) A poset but not totally. (3) An equivalent relation.
8. Consider the set $X = \{a, b, c\}$, and the relations

   $\rho = \{(a, b), (b, b), (b, a)\}$, $\sigma = \{(a, a), (b, b), (c, c), (a, b), (a, c) (b, c)\}$, and $\delta = \{(a, a), (b, b), (c, c), (a, b), (b, a)\}$. Find:

   (1) $\rho \circ \rho$, (2) $\sigma \circ \sigma$, (3) $\delta \circ \delta$, (4) $\rho \circ \sigma$, (5) $\rho \circ \Delta$, (6) $\rho \circ \nabla$.
9. Let **PO**(X) be the collection of all posets on X. Show that (**PO**(X), $\subseteq$) is itself a poset.

10. Let $(A, \leq)$ be a poset and S be a subset of A. If **LUB**(S) exists and **LUB**(S) $\in S$, then show that $(S, \leq)$ has a largest element otherwise $(S, \leq)$ has no a largest element. Discuss a similar result with respect to the **GLB**(S).

11. Find the Hasse Diagram of all possible posets that can be defined on the set $A = \{a, b, c\}$.

12. Find all possible equivalent relations **E**(X) on the set $X = \{x, y, z\}$, and then give the Hasse Diagram of the poset (**E**(X), $\subseteq$).

13. A reflexive, transitive, binary relation is called a ***quasiorder***. Let **R** be a quasiorder on a set X. Define $x \equiv y$ if x **R** y and y **R** x. Prove that $\equiv$ is an equivalence relation, <u>and **R** induces a partially order on X.</u>

14. Give a proof for Theorem 3`.

15. Let $\boldsymbol{G}$ be a group and **S**($\boldsymbol{G}$) be the set of all subgroups of $\boldsymbol{G}$. For $\boldsymbol{S_1}$ and $\boldsymbol{S_2} \in$ **S**($\boldsymbol{G}$), define $\boldsymbol{S_1} \leq \boldsymbol{S_2}$ by the meaning that $\boldsymbol{S_1}$ is a subgroup of $\boldsymbol{S_2}$. Show that (**S**($\boldsymbol{G}$), $\leq$) is a poset and then prove that it is also strictly inductively ordered. Find all maximal elements of (**S**($\boldsymbol{G}$), $\leq$).

16. Find a poset having only one maximal element, but it is not the largest element.

17. Prove that **GLB**(x, y, z) = **GLB**(**GLB**(x, y), z) in a poset $(A, \leq)$?

18. A relation ρ is antisymmetric on $X \Leftrightarrow \rho \cap \rho^{-1} \subseteq \Delta$..